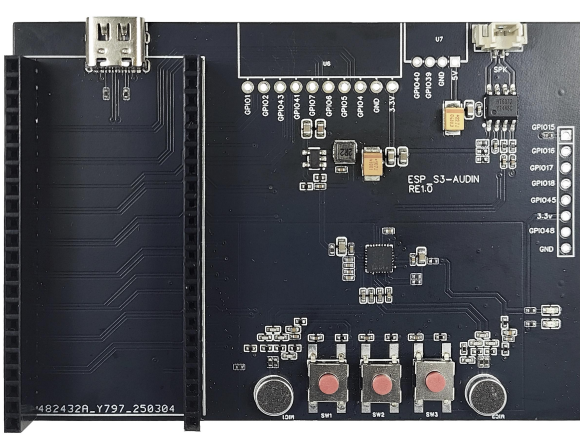
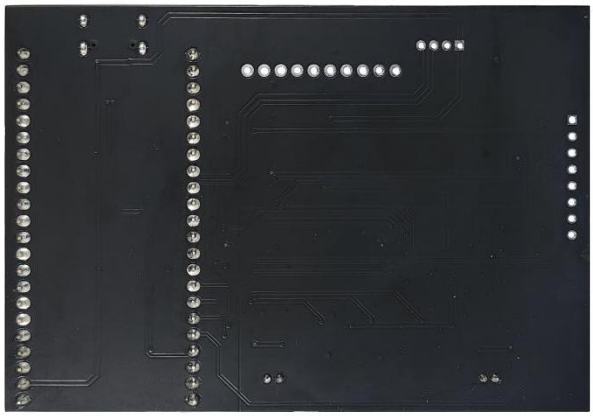
N16R8 Expansion Board Module products specification

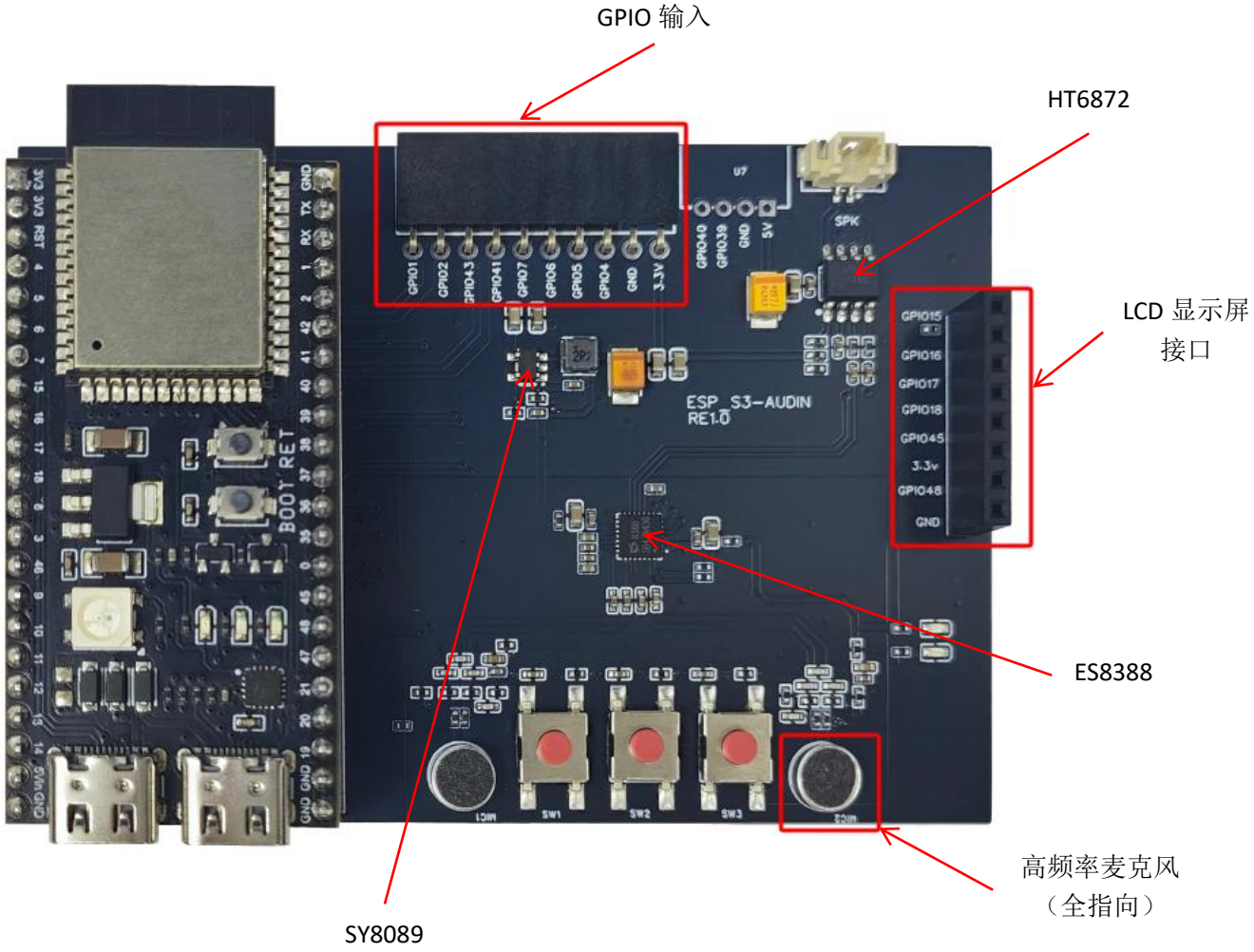




Features:

The N16R8 expansion board is a high-performance development board based on the ESP32-S3R8 chip, specifically designed for audio acquisition and processing. It fully exploits the powerful features of the ESP32-S3, including high-performance I2S interfaces and ADCs, supporting the connection of external digital microphones (such as MEMS microphones) and analog microphones, capable of achieving a high sampling rate of up to 192kHz and multi-channel audio input. The expansion board is equipped with built-in audio gain control and filtering functions, ensuring high-quality and clear audio signals. Additionally, the N16R8 expansion board supports multiple development environments, such as ESP-IDF and Arduino IDE, and provides rich library functions and sample codes, significantly reducing the development difficulty. Its high performance and ease of use make it highly suitable for applications such as voice recognition, recording devices, and smart home systems.

Module hardware layout



Functions Overview

MCU: N16R8 Expansion Board	A high-performance Wi-Fi + Bluetooth dual-mode SoC with an integrated I2S interface and ADC, suitable for audio capture and processing. It features a maximum clock speed of 240 MHz, 512 KB SRAM, and supports high-speed Octal SPI flash and external RAM with larger capacity, making it ideal for complex task processing.
ADC Input	Connects to analog microphones via GPIO pins, supporting the conversion of analog signals to digital signals for processing.

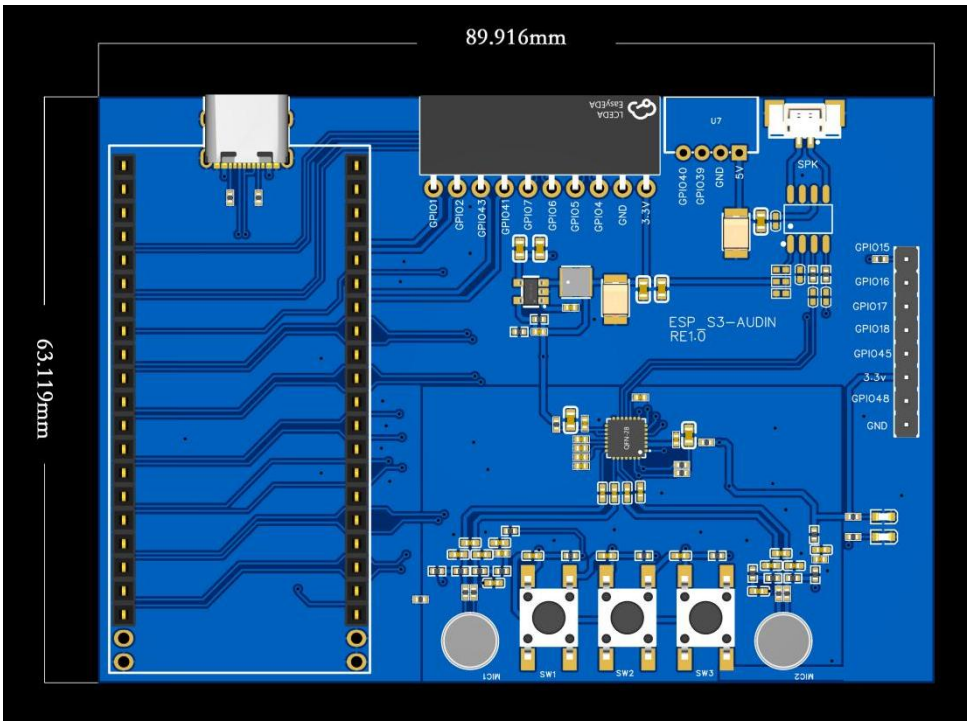
ES8388	Supports 24-bit/192kHz high-resolution audio processing, ensuring clarity and detail in audio signals. It integrates 2-channel ADC, 2-channel DAC, microphone amplifier, headphone amplifier, digital audio effects processing, and analog mixing and gain functions. It also supports multiple audio interfaces such as I2S and PCM, facilitating connection with various microcontrollers and processors.
HT6872	Features low electromagnetic interference (EMI) characteristics, effectively preventing clipping distortion, and has full-bandwidth EMI suppression capabilities, ensuring pure and clear audio quality. HT6872 also includes multiple protection functions, including overcurrent protection, overheat protection, and undervoltage protection, extending the device's lifespan and enhancing user safety.
SY8089A1AAC	Stably operates within an input voltage range of 2.5V to 5.5V, providing up to 2A output current to meet various load requirements. Its high switching frequency of 1.5 MHz ensures rapid response while reducing the size of external filter components. The device integrates a main switch and a synchronous switch, featuring extremely low RDS(ON), significantly reducing conduction losses and improving overall efficiency.

Application scenarios of the module
DIY secondary development
Voice Recognition
Smart Home

Size	
Length	89.916mm
width	63.119mm

Document Update Record

Module size diagram



Unit: mm